

**A REPORT ON INDUSTRIAL TRAINING AT
TRACO CABLE COMPANY, IRUMPANAM**

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INTRODUCTION

TRACO CABLE COMPANY, a Premier Kerala Government Company, commenced operations in the year 1964, manufacturing high quality Electric Cables and Wires in Technical Collaboration with M/s. Kelsey Engineering Co. Ltd., Canada. Since then TRACO has been in the forefront in meeting the needs of Public Sector Undertakings in India like Railways, Electricity Boards of various states in the country and others for AAC/ACSR, Power and Signalling Cables.

One of India's most sought after Paper Insulated Lead Sheathed Telecommunication Cables were produced by TRACO in collaboration with Hindustan Cables, West Bengal under an agreement signed in 1974 until the liberalisation of Licencing policy in the country, TRACO was one of the two manufactures of Telephone Cables in India and the only one in the whole of South India.

With the progress in Cable Technology, Paper Insulated Cables gave way to the much more sophisticated Jelly Filled Telephone cables which are superbly suited for communications. TRACO was one among those who first preceived the opportunities inherent in this new development. It soon went into Technical collaboration with M/s. General Cables Inc. , USA, world leaders in the Communication cable field and manufactured them in India to exacting standards.

Traco launched house wiring cable unit at Thalassery, Kannur district,Kerala.

PRODUCTS

Bare Conductors, Weather-Proof Cables and Power are used by several Electricity Boards in India manufactured to meet the Indian Standard Specification. With stringent quality control methods, cables are of excellent quality and have been acclaimed as one of the best in the country. Signalling Cables made to Railway specifications are functioning all over India. This is because TRACO never compromises on quality.

- **Self Supported Aerial Cable**

Polythene Insulated Moisture barrier self supporting aerial cable. Suspension wire and the cable core are contained in a single polythene sheath 10 pairs with 0.4, 0.5, 0.63 and 0.9 mm gauge. Spec:
Telecommunication
Research Centre Spec. No. 52-23/85 TRC with reference to various
ASTM, BS and IS specification

- **PVC Drop Wire**

PVC insulated drop wire Spec: ITD Spec. No.S/WS-118D & P.V.C.
insulated and polythene sheathed aerial cables 5 pairs/10 pairs
Spec. No. SP/76-2/87/6

- **Bare Conductors**

Bare Conductors - AAC, ACSR and all Aluminium Alloy conductors for transmission and distribution purposes: Spec: IS 398 Part I IS 398
Part II
IS 398 Part IV

- **Flat Twin Cable**

Flat Twin weather proof cables - PVC insulated Aluminium Flat Twin Cables:

1100 V. Grade Spec: IS 694 Extensively used for providing connections to the domestic consumer.

- **Power Cable with Aluminium Conductor**

PVC insulated and sheathed armoured/unarmoured Aluminium power cables

1100 V. Grade. Spec: IS : 1554 Part I

- **Power Cables With Copper Conductor**

PVC insulated and sheathed armoured/unarmoured copper power cables standard Specification : IS : 1554 Part I 1100 V. Grade

- **Railway Signalling Cables**

Signaling Cables - Used by Railways. PVC insulated & sheathed armoured/unarmoured Copper cables. 1100 V. Grade.

Spec : Indian Railway Specification No : S63 - 89 for PVC insulated cables for Railway Signalling

- **House Wiring Cable**

High-grade copper insulated by virgin PVC

Spec : 1 sq.mm , 1.5 sq.mm , 2.5 sq.mm , 4 sq.mm

MANAGEMENT OF THE COMPANY

TRACO Cable Company is registered under the Indian companies act 1956. The registered office of TRACO cable is in Panampilly Nagar,Ernakulam. The management of the company is vested with Board of directors.

QUALITY POLICY

TRACO Cable Company shall strive for continuous improvement in performance, by meeting the needs of internal and external customers complying with regulations through the involvement of all its employees.

ENVIRONMENT POLICY

TRACO Cable Company shall strive for continuous improvement in its performance in preventing environment pollution, to conserve resources and comply with relevant environmental regulations.

CERTIFICATIONS

By RINA-

ISO 9000 - Quality management

ISO 14000 - Environmental management

ISI Certification mark

QUALITY ASSURANCE DEPARTMENT

TRACO is known for its high quality products. Quality Assurance Department ensures that the production International Standard specifications.

As it concentrates more on quality, the products price tend to rise, when compared to its competitors. If it does not produce the right quality, it may lose its ISO Certification. So care is taken to produce high quality products. Therefore, it is the responsibility of the Quality Assurance Department to check whether the quality of the finished goods match the International standard.

In the department, the quality of the product is made sure to match the International Quality Standards.

QUALITY POLICY OF THE COMPANY -

'TRACO Cable Company Limited shall strive for continuous improvement in performance, by meeting the needs of internal and external customers complying with regulations through the involvement of all its employees.'

The following are the functions of the Quality Assurance Department :

1. Incoming inspection
2. In process inspection
3. Finished goods inspection

PRODUCTION

ACSR cables(Aluminium conductor steel-reinforced)

Main raw materials for ACSR cables are aluminium and steel.

The major suppliers of raw materials for ACSR cables are:

- National Aluminium Company Limited(NALCO)
- Bharat Aluminium Company Limited(BALCO)
- Hindalco Industries
- Tata Steel
- Aradhya Steel

Aluminium wires are of 9.5 mm diameter so the first step of production of ACSR cables is to pass aluminium wires through a wire drawing machine so that we could get aluminium wire at required size.

The wire passes through a number of dies which reduces the diameter of the wire in small decrements.Oil is required to lubricate and to take away the heat generated in the machine.

The oil is cooled using water cooling and is recirculated.

The aluminium wire of required diameter is obtained by adjusting the dies.These aluminium wires are rewound to a bobbin.

Steel wires come in loose coils,next step for production is winding of steel on a bobbin through a steel winding machine.

Next step of ACSR production is placing these bobbins in

tubular stranders.For a (6+1)ACSR there will be six aluminium bobbins and one steel bobbin place in the strander machine.

The machine will twist the six aluminium wire around the steel wire.

The final product ACSR is wound wooden drums.

Later ACSR cable is transferred to quality assurance for final inspection. After final inspection and testing the tested OK cables are handed over to stores.

TYPES OF ACSR CABLES -

- Squirrel - 6 + 1 (include 6 aluminium wire and 1 steel wiring having diameter of 2.11 mm)
- Rabbit - 6 + 1 (includes 6 aluminium wire and 1 steel wire having diameter of 3.55 mm)
- Raccoon - 6 + 1 (include 6 aluminium wire and 1 steel wire having diameter of 4.09 mm)

Armoured Power cables

The main raw materials are Tinned copper,PVC compounds, Masterbatches and Steel.

PVC compounds & Masterbatches are provided by Kadakia Plastics and Chemicals Pvt Ltd ,Dadra and Nagar Haveli.

The first step is to provide an insulating PVC coating to the tinned copper wires.For this, the PVC granules and the masterbatch for different colour is put into the hopper of the wire coating machine where it is molten.

The wire comes out of the machine which now has a layer of PVC on it,this wire is then passed though a long water channel to cool down the PVC.

It is then wound on bobbins.

The next step is to twist a number of these PVC coated wires into a bundle.

Then the bundle of cables is provided a PVC sheath by again coating it with PVC.

This cable is then sent for armouring,steel wires are twisted around the cable in this process.It provides high strength to the cables.

Again ,another layer of PVC sheathing is provided.

These cables are mainly used for Underground cables.

The main customers for ACSR cables are KSEB and electricity boards of various states and Railway department. In the case of telephone cables main market is BSNL and other telephone companies.

The factory uses natural lighting ,and rooftop turbine ventillators to expel the hot air thereby saving electricity.

The heavy raw materials are trasported around the factory using forklifts.

House wiring cables are manufactured at Kannur unit.

Four sizes of house wiring cables available at factory outlets, corporate office at Kochi and liaison office at Thiruvananthapuram.

MAINTENANCE DEPARTMENT

Maintenance department takes care of maintenance and preservation of machinery and infrastructure. It is the responsibility of the maintenance department to make sure that the factory premises are clean and all the necessary facilities are available for a good working environment. The infrastructure is maintained and the machinery used for production is to be maintained and repaired whenever necessary. This will result in smooth working of the production process without any disruption.

In TRACO there are two sections in the maintenance department - Mechanical and Electrical

The main objective of is to ensure machine availability by carrying out breakdown and preventive maintenance.

Break Down Maintenance -

Break Down Maintenance is done when any of the machines in the production unit fails to do its particular work or when it malfunctions.

Preventive Maintenance -

Preventive Maintenance is a precautionary measure that is taken so as to prevent any kind of machine breakdown in the future.

CONCLUSION

This internship gave us an opportunity to improve our technical knowledge and experience.

We were able to understand the manufacturing process of different kinds of electrical conductors and cables , and working of various departments of Traco Cable Company.